

Causal-Slash: Streaming Payments for AI Agents

One deposit on Base. Unlimited vendors. Zero gas.

Execution · Wire speed

- Direct P2P micro-cheques over QUIC/TCP sockets.
- 3.43 μs CPU verification, zero intermediate gas, zero block wait.

Settlement · USDC on Base

- One PerformanceCollateralVault contract holds the surety bond.
- 0-second margin release recycles collateral across vendors instantly.

Security · Game-theoretic

- Double-signing a sequence height algebraically leaks the private key in 20 μs.
- Automatic on-chain bond foreclosure makes default strictly negative-EV ($ROI \leq -95\%$).

SYSTEM BLUEPRINT · 5-NODE FLOW

N1 · Payer

AI agent session signer ·
EOTS key engine

N2 · Channel

Bilateral credit channel ·
\$0.50 exposure cap

N3 · Payee

LLM inference · DePIN
GPU compute

N4 · Detection

P2P watchers & finders

N5 · Enforcement

PerformanceCollateralVa
ult.sol on Base

The Problem: Paying 50 APIs Kills Agent Autonomy

Even a simple 8-service report freezes \$150 in accounts to spend \$0.50

\$0.50

Actual compute used across 8 services

\$150+

Dead capital frozen in minimum deposits

95%

Of autonomous bot requests bounce at card checkout

8 Micro-Services · Real-world workflow

- Search (Tavily/Brave), Scraping (Firecrawl), Fast LLM (Groq), Deep Analysis (DeepSeek), Memory (Pinecone), Code Sandbox (E2B), DePIN GPU (Akash), Charts & Images (Replicate).
- Total actual compute used: **\$0.50**.

Human Friction & The Result

- 8 accounts, each bound to a personal credit card.
- \$10–\$20 minimum per platform = **\$150+ dead capital**.
- Bots have no passports or cards — no autonomy.
- Result: centralized hubs with 100% custody risk.

Market & Incentives: Why Independent Hosts and Agents Adopt This

Unlocking machine commerce for autonomous agents and DePIN compute nodes

Why AI Agents Win · Autonomy

- **1 Deposit on Base:** A single \$10 USDC bond replaces 50 separate credit card accounts.
- **On-The-Fly Access:** Hire any DePIN GPU node or API tool instantly without human onboarding.
- **Zero Stranded Capital:** Pay only for compute consumed, reclaim change in 0 seconds.

Why DePIN & Hosts Win · New revenue

- **Zero Bounce Rate:** Monetize autonomous bots that bounce on credit card forms today.
- **0% Fraud & Chargebacks:** Irrevocable USDC settlements backed by on-chain collateral.
- **No US Entity Required:** Independent GPU clusters (Akash, io.net) get instant global USDC monetization.

Protocol Economics · Realistic

- **0 bps Genesis Fee:** 0% protocol fee at launch to bootstrap adoption.
- **Hard-Capped Governance Hook:** Up to 25 bps (0.25%) settlement fee in contract for future DAO sustainability.
- 40% of forfeited malicious bonds routed to the DAO Treasury. Zero capital burned to Oxdead.

System Architecture: From Sockets to Smart Contract

P2P streaming off-chain, final settlement on Base L2

1. Deposit Once On-chain · Base

- Agent deposits \$10 USDC into PerformanceCollateralVault.sol once.
- One bond acts as universal surety covering all connected vendors simultaneously.



2. Stream P2P Off-chain · wire speed

- Agent streams signed micro-cheques over direct QUIC/TCP sockets as tokens arrive.
- Verified locally in 3.43 μ s on the vendor's CPU. Zero gas, zero block latency.



3. Cooperative Close 0 seconds

- Session concludes: vendor and agent sign a mutual net-receipt.
- Remaining collateral is unlocked in 0 seconds to reuse for the next task.

Security: Why Default Is Mathematically Unprofitable

Closed-loop liquidation ensures 100% vendor default protection

1 • Detect

The agent signs two conflicting cheques at sequence height h .



2 • Extract

The vendor algebraically recovers the agent's private key from the two signatures.



3 • Slash

Extracted key verified on Base L2 in ~12.5k gas to foreclose the bond.

$$sk = (s_1 - s_2) \cdot (e_1 - e_2)^{-1} \bmod q$$

The shared nonce cancels out: one formula, $O(1)$ proof.

Liquidation Waterfall • Guaranteed priority

Priority 1 • Whistleblower / detecting node

15% finder bounty

Priority 2 • Unpaid vendor, from the remaining 85%

100% damage restitution

Priority 3 • Remainder

60% Insurance Reserve

40% Treasury

Negative-EV Guarantee • Unprofitable

- Attack gain: strictly capped at \$0.50 buffer per vendor.
- Forfeited collateral: \$10.00 locked bond on Base.
- Expected ROI $\leq -95\%$. 100% of funds stay on Base.

Benchmarks: Tested on Real Hardware

3.4 μ s on CPU, 70k cheques/sec, 14/14 Foundry tests passed

CPU VERIFICATION

3.43 μ s

End-to-end sign + verify on native C11 engine

SOCKET RTT

14.17 μ s

Real TCP loopback round trip · 70,559 cheques/sec

COST REDUCTION

-99.4%

Vs sending raw L2 transactions

FOUNDRY TESTS

14/14

Invariant & fuzz tests passing · 256 random runs each

Production Daemon · Native C11

- POSIX C11, GCC -O3, stdatomic.h with ring-buffer bitmask indexing.
- Zero heap allocations on hot path. Clean ASan & UBSan (zero leaks, zero overflows).
- Python C-FFI wrapper (libcausal_slash.so) ready for agent pipelines.

Verified On-Chain · Base Sepolia

- PerformanceCollateralVault.sol verified on BaseScan (0xdC68e06331aF5aC885E5A7Cb875d364dedcD1D34).
- 14/14 passing Foundry invariant & fuzz tests (256 random runs each).

Base Ecosystem Alignment: The Sticky TVL Flywheel

Making Base the clearinghouse for autonomous machine commerce

Sticky USDC TVL · Bonds locked on Base

\$50K

Locked Base TVL at 1,000 agents

\$5M

Locked USDC TVL at 100,000 agents

Every active agent locks a \$10–\$50 surety bond on Base.

Mempool Defense · Micro-calls off-chain

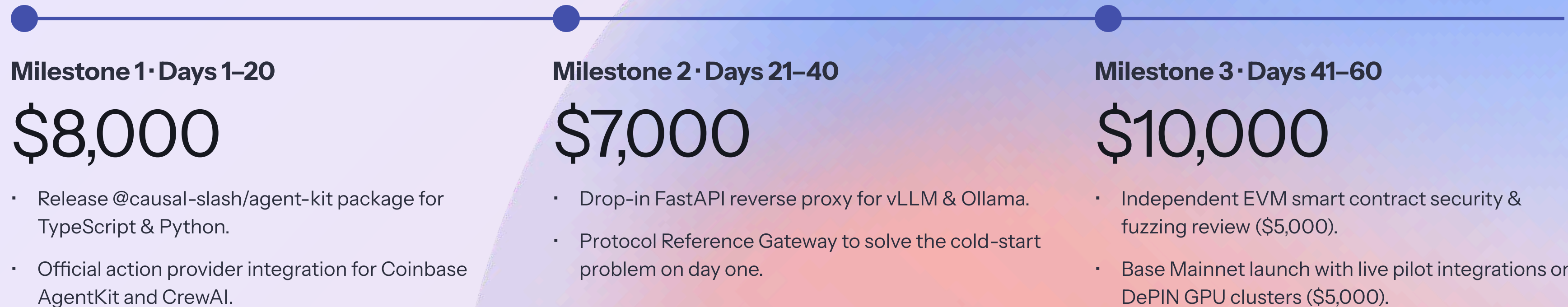
- Millions of micro-transactions stay off-chain over direct P2P sockets.
- Only high-value bonding and net-settlements land on the Base sequencer.

Coinbase AgentKit Native · Drop-in integration

- Turnkey Action Provider for Coinbase AgentKit: agents stream payments in 3 lines of code.
- Universal Drop-In Proxy: wrap any OpenAI-compatible base URL with zero agent code changes.

The Ask: \$25k to Launch on Base Mainnet

60-day sprint from Sepolia testnet to live pilot integrations



\$25,000

Base Ecosystem Builder Grant • 60 days from Sepolia testnet to live pilot integrations on Base Mainnet